can gather enough statistics to convince ourselves that the probability in quantum mechanics really is just the wave function squared.

Many-Worlds is a different story. Say we put an electron into an equal superposition of spin-up and spin-down, measure its spin, then repeat a large number of times. At every measurement, the wave function branches into a world with a spin-up result and one with a spin-down. Imagine that we record our results, labeling spin-up as “0” and spin-down as “1.” After fifty measurements, there will be a world where the record looks like

10101011111011001011001010100011101100011101000001.

That seems random enough, and to obey the proper statistics: there are twenty-four 0’s, and twenty-six 1’s. Not exactly fifty-fifty, but as close as we should expect.

But there will also be a world where every measurement returned spin-up, so that the record was just a list of fifty 0’s. And a world where all the spins were observed to be down, so the record was a list of fifty 1’s. And every other possible string of 0’s and 1’s. If Everett is right, there is a 100 percent probability that each possibility is realized in some particular world.

In fact, I’ll make a confession: there really are such worlds. The random-looking string above wasn’t something I made up to look random, nor was it created by a classical random-number generator. It was actually created by a *quantum* random-number generator: a gizmo that makes quantum measurements and uses them to generate random sequences of 0’s and 1’s. According to Many-Worlds, when I generated that random number, the universe split into 2^{50} copies (that’s 1,125,899,906,842,624, or approximately 1 quadrillion), each of which carries a slightly different number.